Enhanced DTF-AT Audio Event Classification

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Abstract

This report presents a systematic study of the Decoupled Time Frequency Audio Transformer (DTF-AT) architecture applied to large scale audio event classification on the AudioSet balanced subset. The baseline DTF-AT model was successfully reproduced across three independent training runs, achieving a mean Average Precision (mAP) of 0.267, which closely matches the 0.278 mAP reported in the original paper. Building on this foundation, four enhancements were investigated: Time Reversed Augmentation, a Hybrid Pre Stem convolutional block, an Attention Pooling head, and a Multi Scale Fusion head. Time Reversed Augmentation and the Hybrid Pre Stem delivered strong performance gains, with the Hybrid Pre Stem reaching 0.34 mAP under standard single stage training. However, the more structurally complex modifications suffered from severe gradient instability caused by a fundamental conflict between randomly initialized components and ImageNet pretrained weights. To address this, a two stage training strategy was proposed and validated, where the pretrained backbone is first frozen to allow new components to stabilize before joint fine tuning. This strategy pushed the best result to 0.35 mAP and significantly recovered performance across all other configurations. The findings highlight that optimization strategy is just as critical as architectural design when integrating new components into pretrained transformer backbones.

1. Introduction

1.1 Motivation

Audio event classification is a fundamentally challenging problem due to the rich and complex nature of real world acoustic signals. A 10 second audio clip can contain multiple overlapping events such as speech, music, and environmental sounds, all occurring simultaneously across different frequency ranges and time spans. Capturing this complexity requires models that can jointly reason about fine grained spectral details and long range temporal dependencies. Standard convolutional neural networks, while effective at extracting local features, often struggle to model global context across an entire audio clip. On the other hand, standard vision transformers treat audio spectrograms as flat 2D images, which does not fully exploit the distinct physical properties of the time and frequency axes in audio signals. The DTF-AT architecture addresses this gap by decoupling the processing of temporal and spectral dimensions through hierarchical attention mechanisms, making it a well motivated and promising candidate for large scale audio event classification tasks.

1.2 Objectives

The primary objective of this project is to reproduce the DTF-AT baseline on the AudioSet balanced subset and establish a reliable performance benchmark. Beyond reproduction, the project investigates whether targeted architectural modifications and data augmentation strategies can meaningfully improve classification performance. Specifically, we explore whether introducing convolutional preprocessing before the transformer stem can help the model extract richer low level spectro temporal features, and whether replacing standard pooling with learned attention mechanisms can improve the model's ability to localize and identify audio events within a clip. A secondary objective that emerged during experimentation was understanding the optimization challenges that arise when combining randomly initialized components with pretrained transformer backbones, and proposing a principled solution to address them.

1.3 Literature Review

The DTF-AT architecture was proposed by Alex et al. (2024) as a specialized transformer designed for audio spectrogram understanding. Unlike standard vision transformers that apply uniform patch based attention across both spatial dimensions, DTF-AT processes time and frequency axes through separate convolutional paths within its stem block before integrating them via self attention. The encoder follows a hierarchical 2-2-5-2 block configuration across four stages, progressively compressing the input representation while increasing channel depth. This design draws inspiration from hierarchical vision transformers such as Swin Transformer, but adapts the windowed and grid attention mechanisms specifically for the time frequency structure of audio signals.

The AudioSet dataset, introduced by Gemmeke et al. (2017), is one of the largest publicly available audio classification benchmarks, containing over 2 million 10 second clips spanning 527 audio event categories. The balanced subset used in this project contains approximately 20,000 training clips and serves as a standard benchmark for evaluating audio classification models. Prior work on AudioSet includes the Pretrained Audio Neural Networks (PANN) framework and the Audio Spectrogram Transformer (AST), both of which demonstrated that cross domain transfer learning from ImageNet pretrained weights can significantly benefit audio classification performance. Our work builds on these foundations by evaluating the DTF-AT architecture and investigating further enhancements to push its performance on this benchmark.

3. Dataset

3.1 Dataset Description

The dataset used in this project is the AudioSet balanced subset, a large scale multi label audio classification benchmark introduced by Gemmeke et al. (2017). The processed dataset consists of approximately 12 GB of training data and 10 GB of test and evaluation data. The training split contains 20,550 audio clips while the test and evaluation split contains 18,886 clips. All audio is standardized to a sampling rate of 32 kHz with a fixed duration of 10 seconds per clip, ensuring a consistent input format across the entire pipeline. The dataset supports a 527 class label space as defined in the provided CSV label file, which serves as the complete vocabulary for both training and evaluation. Each clip can carry multiple labels simultaneously, making this a multi label classification problem where overlapping acoustic events are common within a single recording.

3.2 Dataset Preprocessing and Data Cleaning

Beyond corruption handling, the raw JSON provides labels as plain strings while the CSV label file uses standardized class identifiers known as MID codes. To ensure consistency throughout the training and evaluation pipeline, each string label in the JSON was mapped to its corresponding MID code using the CSV vocabulary. All processed clips were then saved using these stable class identifiers. As a qualitative sanity check on the preprocessing pipeline, the DTF-AT encoder output was visualized for a representative sample clip with labels Sniff and Speech. The resulting time frequency activation map showed structured, non uniform activations across specific frequency bands and time regions, confirming that the input pipeline is functioning correctly and that the encoder is responding meaningfully to audio content rather than producing arbitrary activations.

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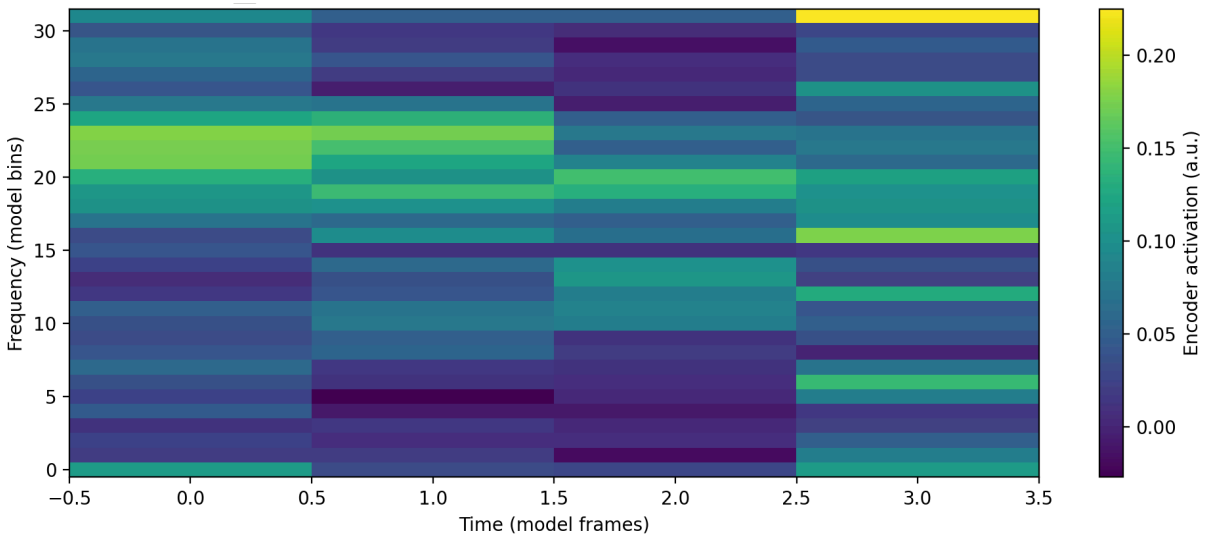

Fig-1: TF Encoder Output for Audio Sample

3.3 Exploratory Data Analysis

Two key analyses were conducted to better understand the structure and characteristics of the dataset before training.

The first analysis examined label cardinality, which refers to the number of labels assigned to each clip. Since AudioSet contains overlapping acoustic events, a single 10 second clip can carry multiple simultaneous labels. The distribution across both the training and test splits shows that most clips carry approximately 2 to 3 labels, with the distribution peaking around 1 to 3 labels and tapering off steadily as label count increases. Importantly, no unlabeled clips were found in either split, confirming that the dataset is well formed and ready for multi label training with independent class probabilities and a binary cross entropy loss objective.

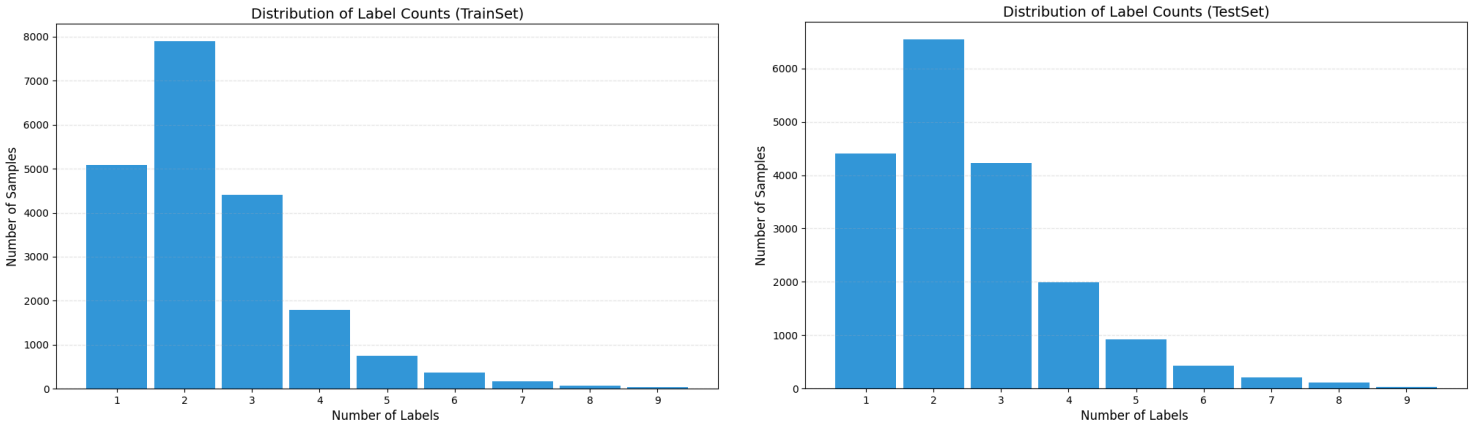

Fig-2: Label Cardinality Distribution Per Audio

The second analysis examined label co-occurrence through a Jaccard Similarity matrix computed over the top 10 most frequent labels in the dataset. The heatmap reveals clear and interpretable correlations between semantically related classes. For example, Animal and Domestic animals/Pets show strong co-occurrence, as do Guitar, Plucked string instrument, and Musical instrument. The majority of off-diagonal values are close to zero, indicating that most class pairs co-occur rarely and that the label space is largely sparse. These co-occurrence patterns have practical implications for model evaluation, as strong label priors can lead to false positives where the model over-predicts related classes in ambiguous audio mixes. This observation motivates the use of proper multi label evaluation metrics and threshold calibration rather than a single fixed decision boundary.

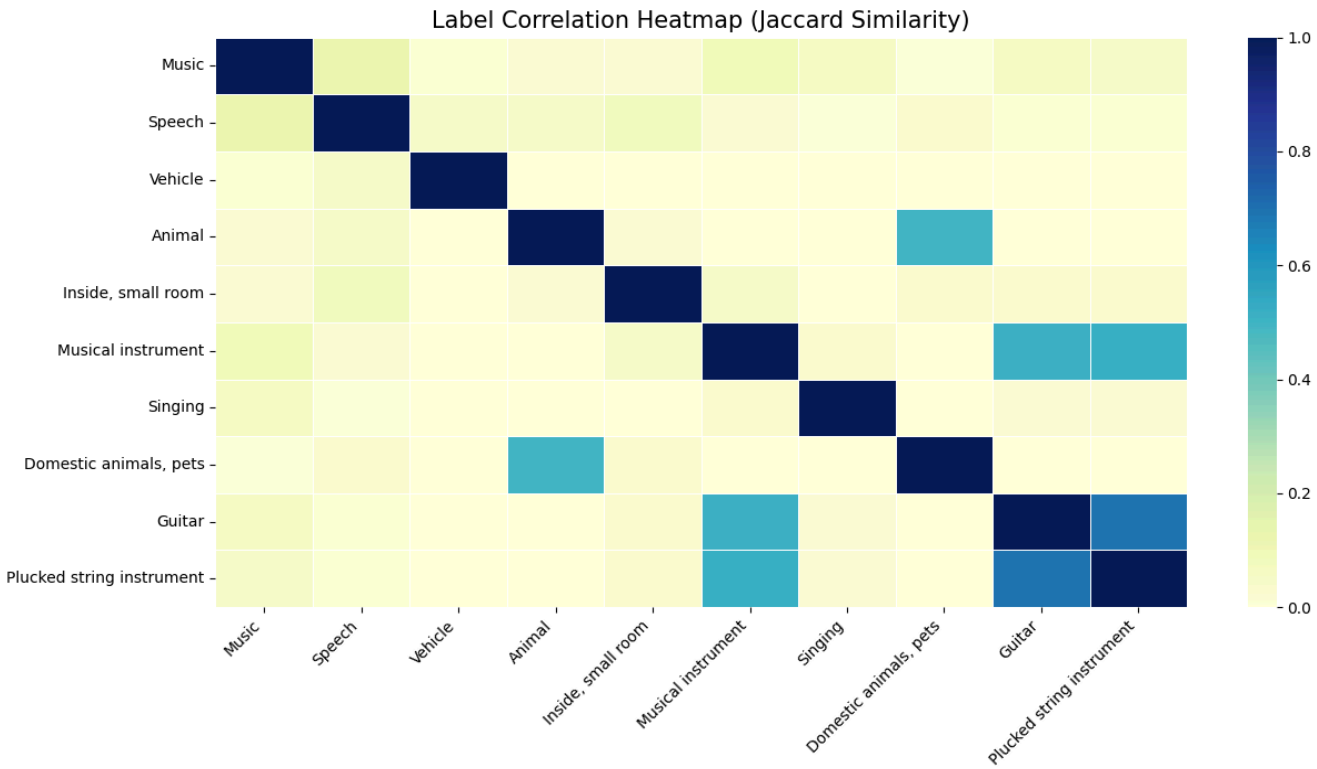

Fig-3: Label Jaccard Similarity Heatmap

4. System Design and Implementation Details

4.1 Baseline Architecture

The baseline system processes 10 second mono audio recordings sampled at 32 kHz. Preprocessing involves generating Kaldi style log mel filterbank features with 128 mel frequency bins and a 10 ms frame shift, producing input spectrograms of shape 1024 x 128 representing time frames by frequency bins respectively.

The DTF-AT encoder consists of a stem block followed by four hierarchical stages arranged in a 2-2-5-2 block configuration. The stem block processes the input spectrogram through two parallel convolutional paths, one operating along the frequency to time direction and the other along the time to frequency direction. These paths are blended through a learned weighted combination controlled by a scalar parameter, producing an initial representation that captures both temporal and spectral local structure before being passed into the transformer stages.

Each stage contains DTF-AT blocks that apply a sequence of operations including a time frequency mobile convolution block, window attention, a feed forward network, grid attention, and another feed forward network. Window attention captures local dependencies within spatial windows while grid attention captures long range

dependencies across the full feature map. This combination allows each block to model both local texture and global context efficiently.

Through the four stages, the input representation is progressively compressed while channel depth increases from 64 in Stage 1 to 96, 192, 384, and finally 768 in Stage 4. The final encoder output has a spatial shape of 32 x 4 representing 32 frequency bins by 4 time bins, which is then passed through adaptive pooling, flattening, and a linear classifier to produce predictions over the 527 class label space. The model is initialized with ImageNet-1K pretrained weights to leverage cross domain transfer learning, following the same approach described in the original paper.

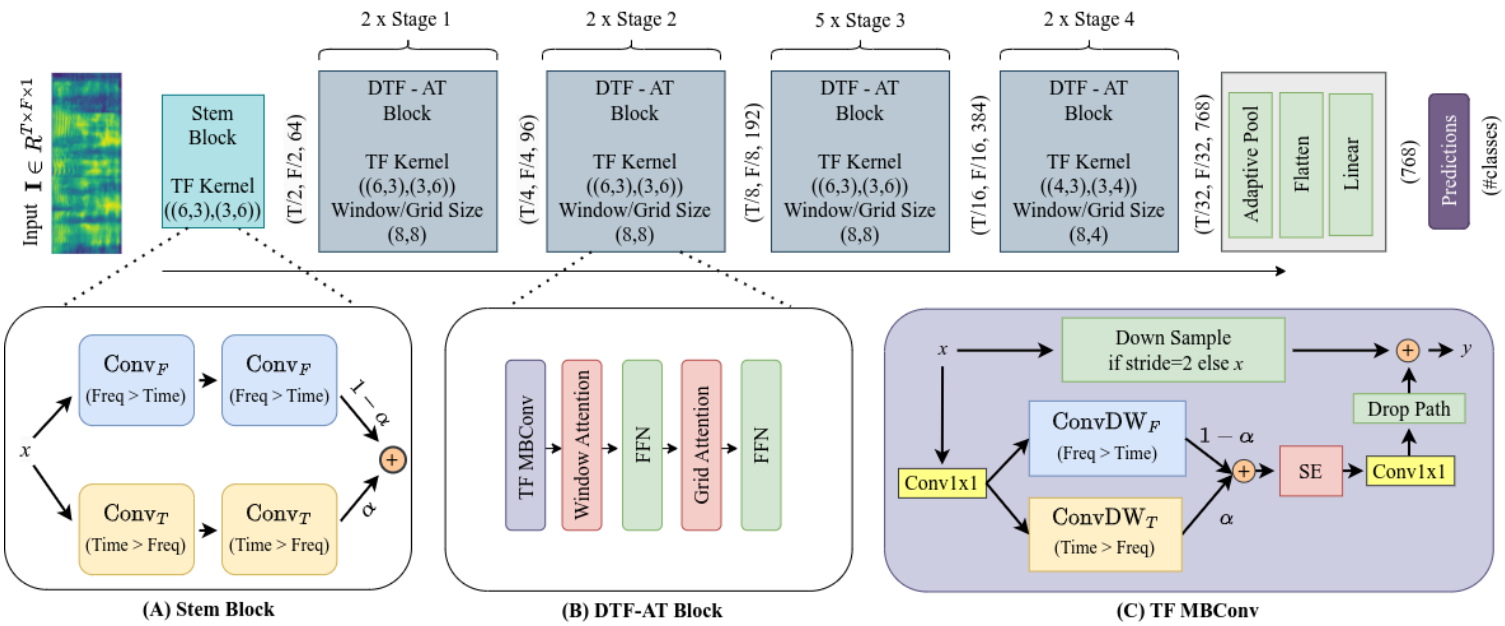

Fig-4: Original DTF-AT architecture from Research Paper.

4.2 Technologies and Tools

The implementation required careful integration of two separate codebases. The DTF-AT repository provided the core model architecture definitions and evaluation metrics, while the Audio Spectrogram Transformer (AST) repository provided the training infrastructure including data loading utilities, training loops, and augmentation strategies. This integration was necessary because the original DTF-AT repository does not include a standalone training pipeline suitable for AudioSet scale experiments.

The model was implemented in PyTorch and trained on dedicated GPU resources. ImageNet-1K pretrained weights were used for model initialization across all experiments. Evaluation was performed using the mean Average Precision metric following the PANN evaluation protocol on the AudioSet balanced evaluation set, which is the standard benchmark used in the original paper for fair comparison.

4.3 Proposed Enhancements

Four enhancements were designed and evaluated on top of the baseline DTF-AT architecture.

The first enhancement is **Time Reversed Augmentation**. This strategy pairs every original 10 second audio clip with a time reversed version under the same label during training, effectively doubling the training data without requiring any additional labeled samples. The underlying assumption is that many audio events such as gunshots, crying, and

instrument strums retain their acoustic identity through their frequency content regardless of temporal direction. By exposing the model to both forward and reversed versions of each clip, the model is encouraged to focus on spectral signatures rather than temporal order, which acts as a regularization effect and improves generalization.

The second enhancement is the **Hybrid Pre Stem**. A residual convolutional block is inserted before the standard transformer stem block. This block applies two sequential 3x3 convolutions with batch normalization and ReLU activations, combined with a residual skip connection that preserves the original input. The output shape is identical to the input shape of (T, F, 1), making the block fully compatible with the existing pretrained stem block downstream. The motivation is that local convolutions are better suited for extracting low level features such as harmonic textures, note onsets, and spectral edges compared to the patch based attention in the transformer stem alone. The residual connection also ensures that even before the convolutional layers have learned anything useful, the original spectrogram information passes through unchanged, protecting the downstream pretrained backbone from disruption.

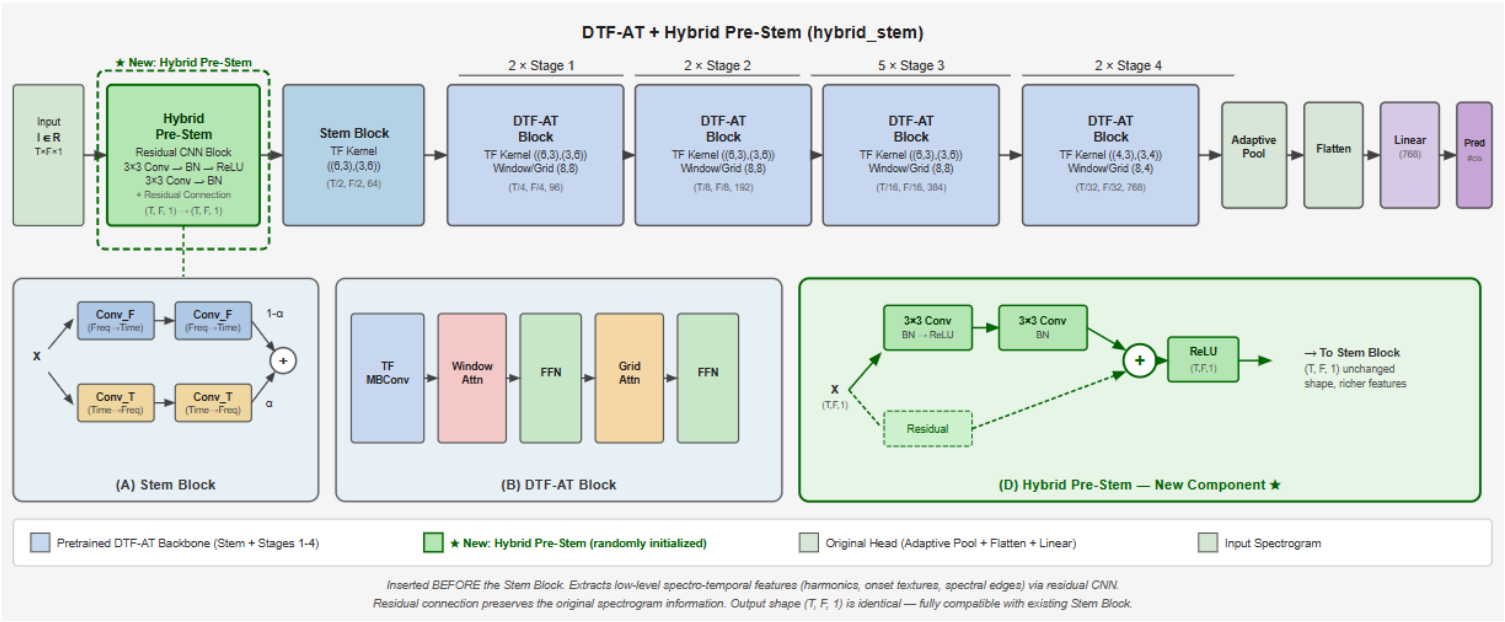

Fig-5: DTF-AT + Hybrid Pre Stem Architecture Diagram

The third enhancement is the **Attention Pooling Head**. The standard adaptive pooling followed by flattening and linear classification is replaced with a multi head attention pooling layer. A learnable query vector of dimension 768 attends over all spatial positions in the final 32 x 4 feature map from Stage 4 using multi head attention with 8 heads and a head dimension of 96. The attended representation is then passed through layer normalization and an MLP before being fed into the linear classifier. This allows the model to dynamically focus on the specific time frequency regions that are most relevant for each prediction rather than treating all spatial positions equally as average pooling does.

The fourth enhancement is the **Multi Scale Fusion Head**. Instead of only using the final Stage 4 representation for classification, this head collects intermediate representations from all four encoder stages. Each stage output is passed through an adaptive pooling layer to project it to a unified 768 dimensional representation. The four projected stage tokens are stacked into a sequence of shape 4 x 768. A cross attention mechanism then uses the Stage 4 token as a query and all four stage tokens as keys and values, allowing the final semantic representation to selectively attend to relevant fine grained information from earlier stages. A learnable weighted sum then combines all four stage representations, and the result is passed to the linear classifier. This design aims to produce a richer feature representation by blending low level spectro temporal details from early stages with high level semantic patterns from the final stage.

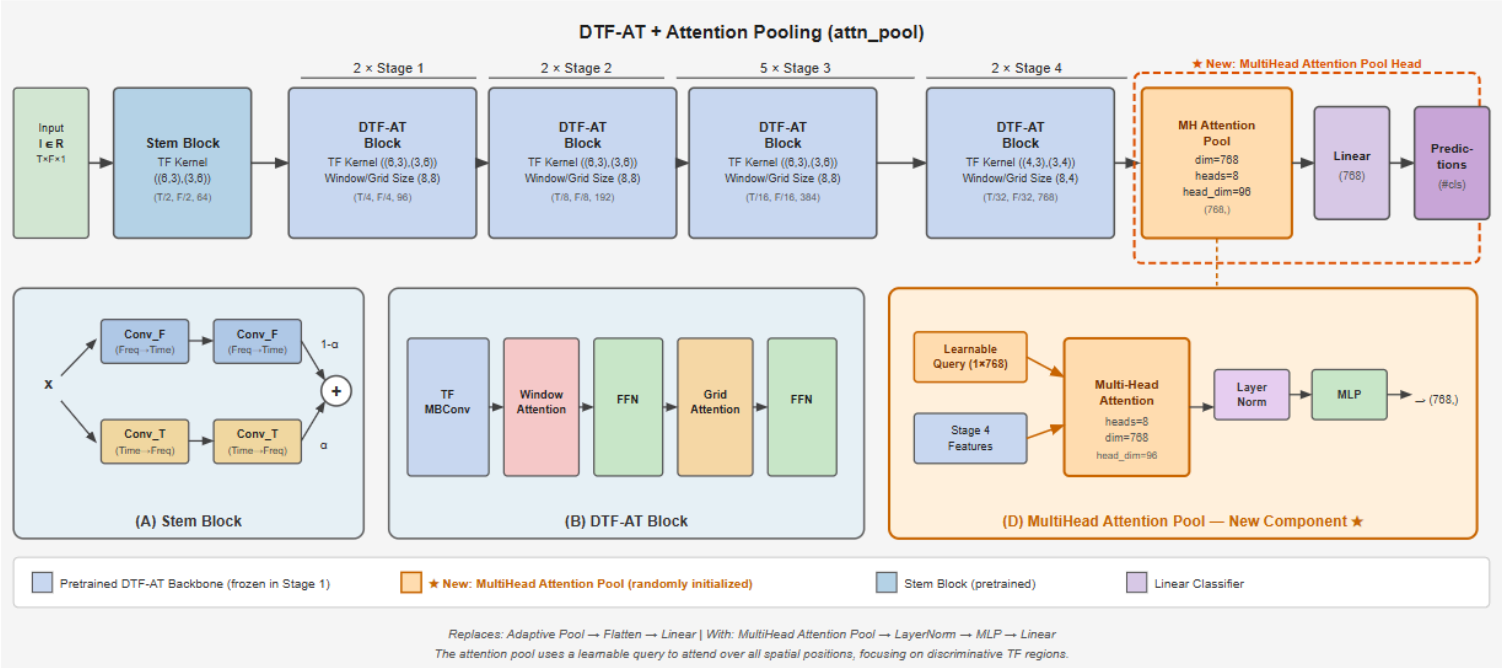

Fig-6: DTF-AT + Attention Pooling Architecture Diagram

Finally, an **All Combined configuration** was also evaluated, which simultaneously integrates all three architectural modifications: the Hybrid Pre Stem at the input, the Multi Scale Fusion Head tapping all four stage outputs, and the Attention Pooling Head operating on Stage 4 features. The outputs of the Multi Scale Fusion and Attention Pooling are summed before being passed to the linear classifier.

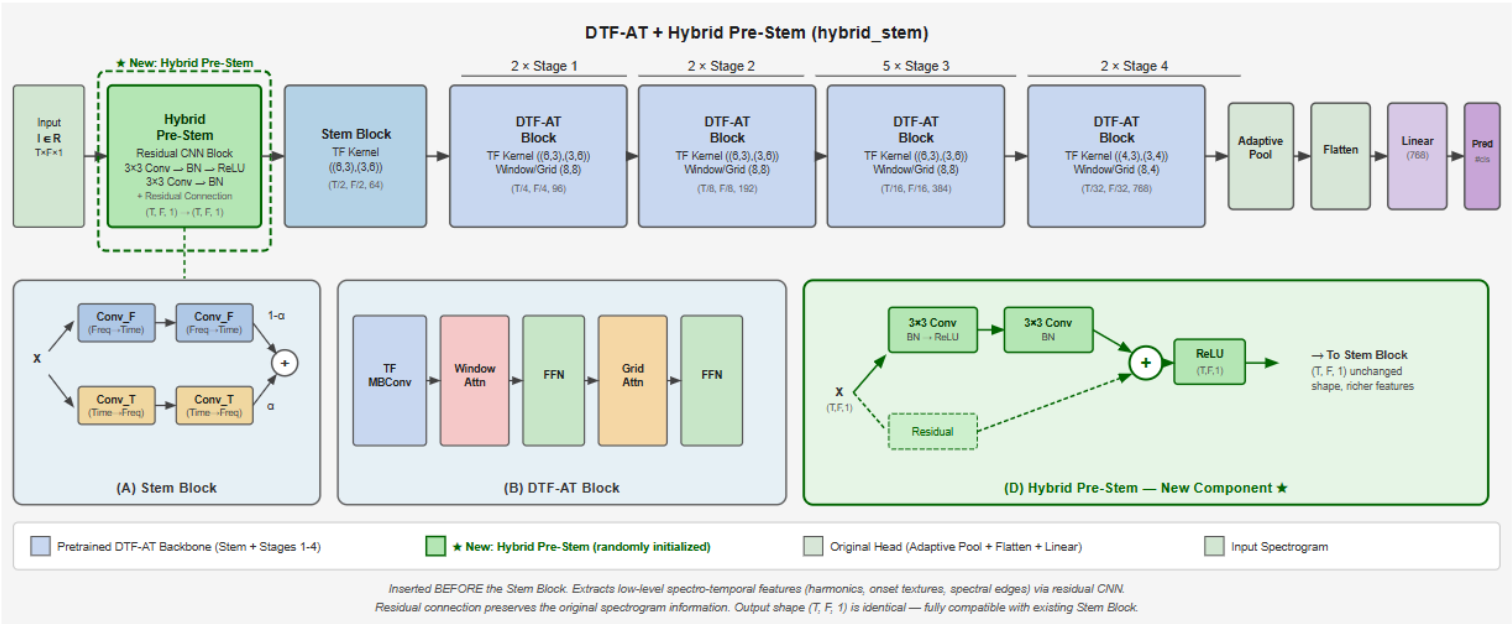

Fig-7: DTF-AT + Multi Scale Fusion Head Architecture Diagram

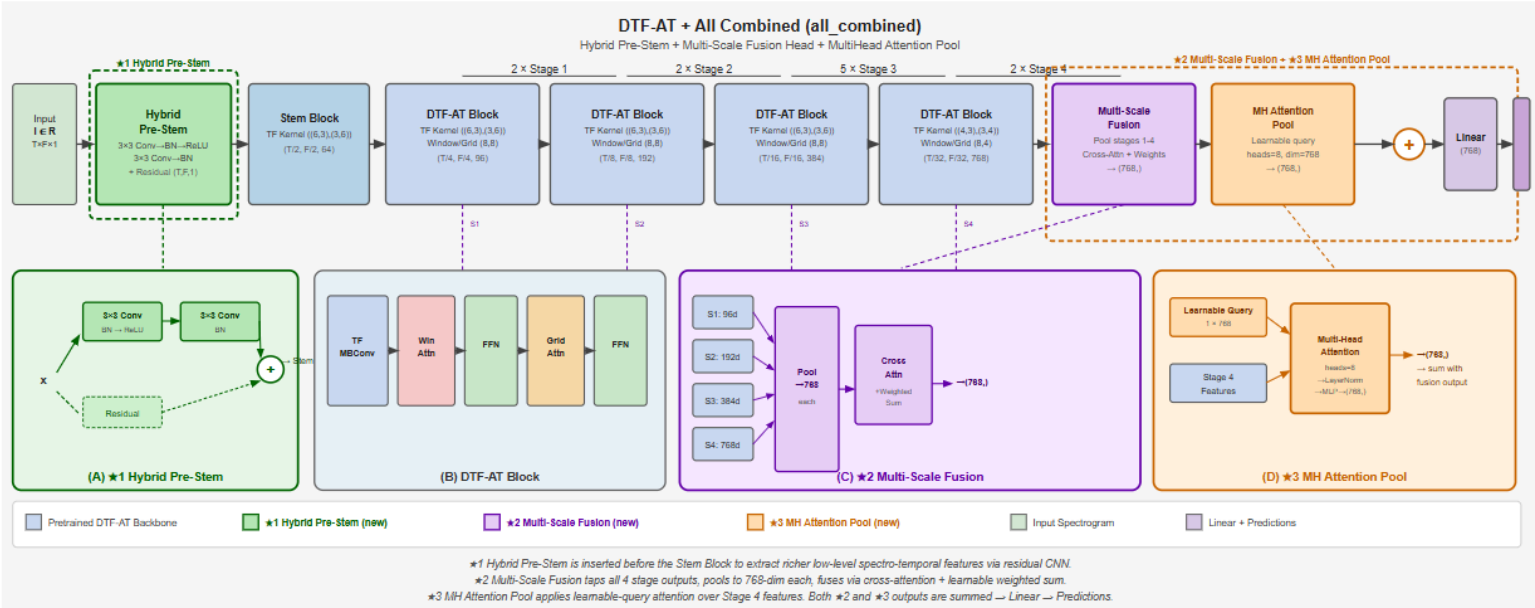

Fig-8: DTF-AT + All Combined Architecture Diagram

4.4 Hyperparameter Configuration

All experiments including the baseline and all enhanced configurations were trained using the same hyperparameter setup to ensure fair and controlled comparison. The AdamW optimizer was used with Binary Cross Entropy loss as the training objective, which is appropriate for the multi label nature of the AudioSet classification task. Training was conducted for 50 epochs with a batch size of 32 and an initial learning rate of 5e-4. A linear warm up was applied over the first 1,000 steps to stabilize early training before the full learning rate took effect. A multi step learning rate scheduler was applied with decay milestones at epochs 10, 15, 20, and 25, reducing the learning rate by a factor of 0.1 at each milestone. This aggressive decay schedule was chosen to match the training protocol described in the original DTF-AT paper for the AudioSet balanced subset.

5. Experiments and Results

5.1 Baseline Reproduction

To establish a statistically reliable baseline, the DTF-AT model was trained three times from scratch using random seeds 0, 1, and 2. All three runs followed identical hyperparameter configurations and data pipelines as described in Section 4.4. Training was conducted for the full 50 epochs without early stopping, consistent with the protocol described in the original paper.

The training dynamics across the three runs showed consistent behavior. The model underwent a rapid learning phase during the first 15 epochs, with mAP rising sharply from near zero to approximately 0.26. This initial phase corresponds to the model adapting its ImageNet pretrained representations toward acoustic feature discrimination. Following the learning rate decay milestones at epochs 10 and 15, the mAP curve plateaued and stabilized, indicating that the model had reached convergence. The learning rate approached negligibly small values by epoch 20, and the remaining 30 epochs served primarily to consolidate the learned representations. This observation suggests that early stopping around epoch 20 could have been implemented to conserve GPU resources without any meaningful sacrifice in final performance.

The training and validation loss curves revealed an interesting characteristic of the optimization process. The training loss remained extremely low throughout, hovering between 0.01 and 0.02, while the validation loss stayed relatively constant at approximately 0.70. This large gap is not indicative of overfitting but rather reflects a fundamental difference between the two metrics. Binary Cross Entropy loss is sensitive to absolute predicted probability values, while mAP is a ranking based metric that focuses on the relative ordering of classes. The sustained improvement in mAP despite the high validation loss confirms that the model successfully learned to rank correct classes higher even when its absolute probability estimates were not perfectly calibrated.

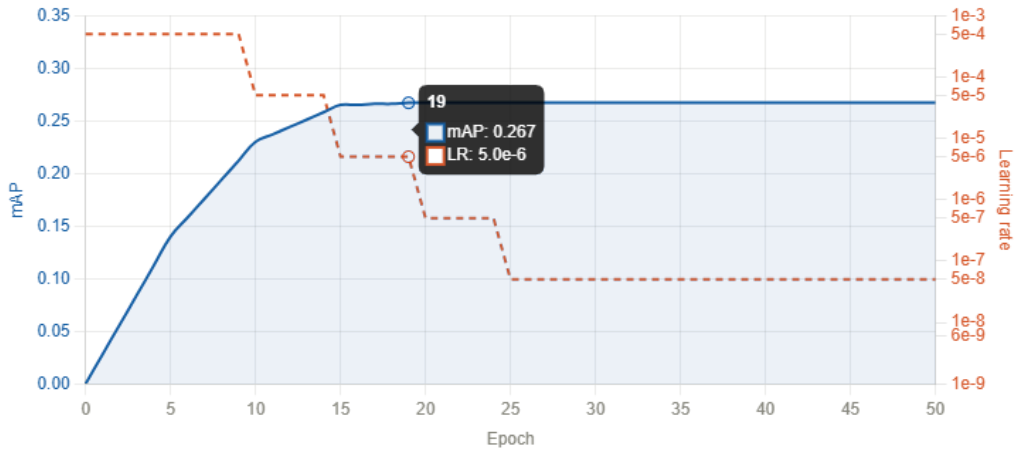

Fig-9: Training Dynamics - mAP Progression and Learning Rate over 50 epochs

Across three seeds, the model achieved a mean mAP of 0.267 on the PANN balanced AudioSet evaluation set, closely matching the 0.278 mAP reported in the original paper. This small gap is expected given natural variation in hardware environments and random initialization, and the result confirms a successful and faithful reproduction of the DTF-AT baseline.

5.2 Encoder Representations Analysis

Prior to running enhancement experiments, a statistical analysis of the 32 x 4 encoder representations was conducted across multiple audio samples to verify that the baseline model was producing meaningful and discriminative features. The analysis examined the activation properties of the encoder output and revealed the following ranges across samples: minimum activations between 0.02 and 0.04, maximum activations between 0.20 and 0.27, and mean activations between 0.07 and 0.09.

More importantly, pairwise cosine similarity analysis across different audio samples showed that these representations were not uniform. The cross sample variability confirmed that the encoder was capturing discriminative structure that varied meaningfully across different audio inputs rather than collapsing to a common generic template. This analysis validated that the 32 x 4 feature map carries sufficient information for downstream classification and that the baseline model is a solid foundation for the proposed enhancements.

5.3 Ablation Studies

A systematic ablation study was conducted to evaluate the impact of each proposed enhancement individually as well as in combination. Each configuration was evaluated using the same training setup and reported in terms of mAP on the AudioSet balanced evaluation set. Two training regimes were tested where applicable: single stage training where all components are trained jointly from the start, and two stage training where the pretrained backbone is first frozen for the initial 10 epochs to allow new randomly initialized components to stabilize and learn the data distribution, followed by 40 epochs of full joint fine tuning where the entire network is unfrozen and trained end to end.

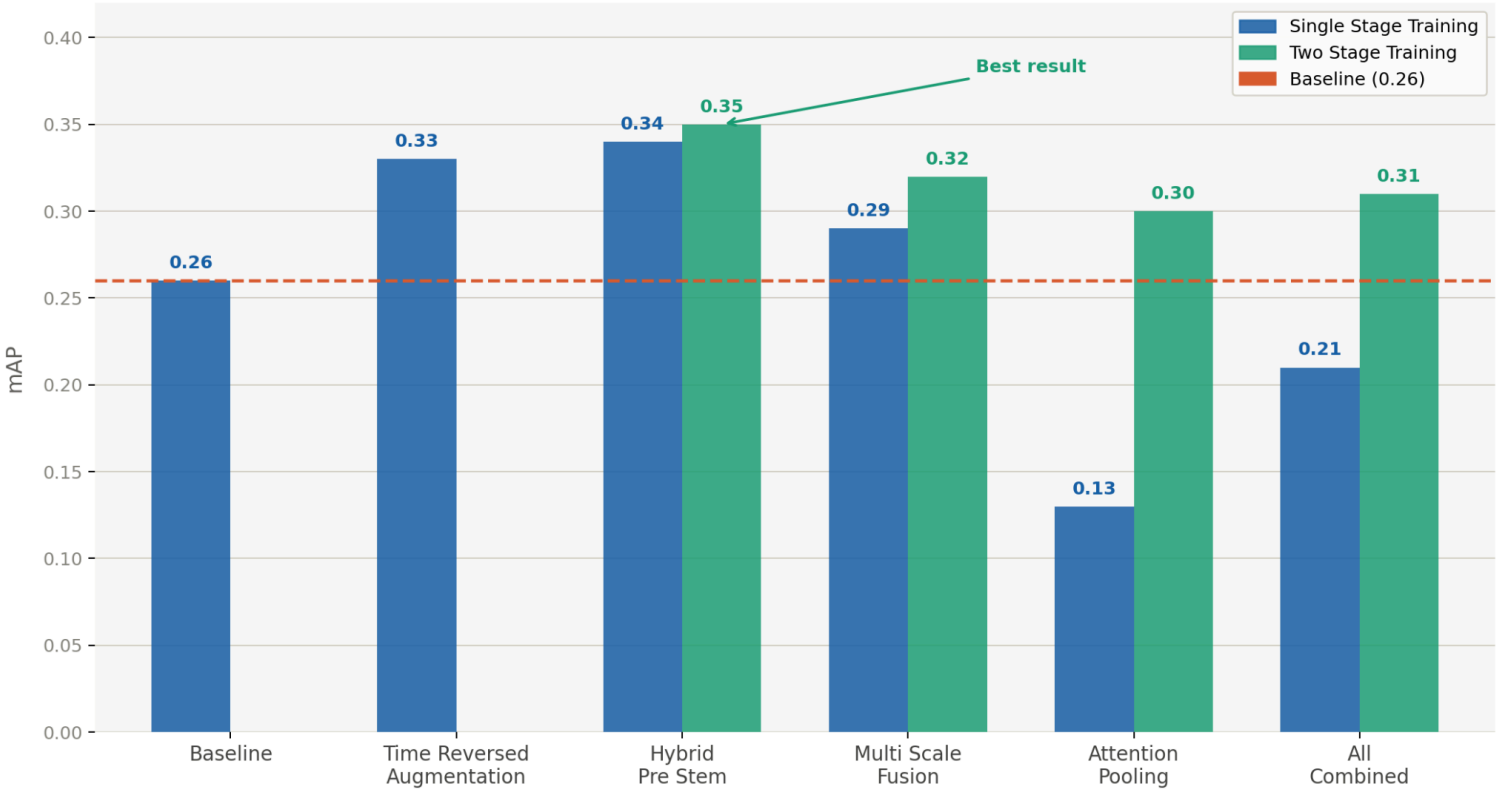

Fig-10: Ablation Study Results mAP Comparison Across Configurations for SIngle & Two Stage Trainings

The results demonstrate that meaningful performance gains can be achieved through both smarter data usage and targeted architectural modifications. Time Reversed Augmentation provided a 0.06 mAP gain by encouraging the model to focus on spectral signatures rather than temporal order, while the Hybrid Pre Stem achieved the best overall result of 0.35 mAP, showing that enriching low level features before the transformer stem through a simple residual block can be more effective than adding complexity at the classification head. The more structurally complex modifications suffered from severe gradient instability under single stage training, caused by a fundamental conflict between randomly initialized components and well tuned ImageNet pretrained weights. The proposed two stage training strategy resolved this by freezing the backbone for the first 10 epochs before full joint fine tuning, recovering the Attention Pooling Head from 0.13 to 0.30 mAP and the All Combined configuration from 0.21 to 0.31 mAP, demonstrating that optimization strategy is just as critical as architectural design when integrating new components into pretrained transformer backbones.

6. Conclusion

This project successfully reproduced the DTF-AT baseline on the AudioSet balanced subset, achieving 0.267 mAP across three independent training runs, closely matching the 0.278 mAP reported in the original paper. Building on this foundation, four enhancements were investigated, with the Hybrid Pre Stem and Time Reversed Augmentation proving to be the most effective, reaching a best result of 0.35 mAP. A key finding of this study is the gradient instability that arises when integrating randomly initialized components with pretrained transformer backbones, and the two stage training strategy proposed to address it proved highly effective across all complex configurations. Overall, the results highlight that thoughtful optimization strategy combined with architecturally compatible enhancements can push the performance of state of the art audio transformers meaningfully beyond their reported baselines. All source code, model configurations, and training scripts for every experiment conducted in this study are provided in our GitHub repository for reproducibility and further exploration.

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